Multimodal NGS analysis of gene expression and chromatin state in the olfactory epithelium

UBDS^3 | 2026

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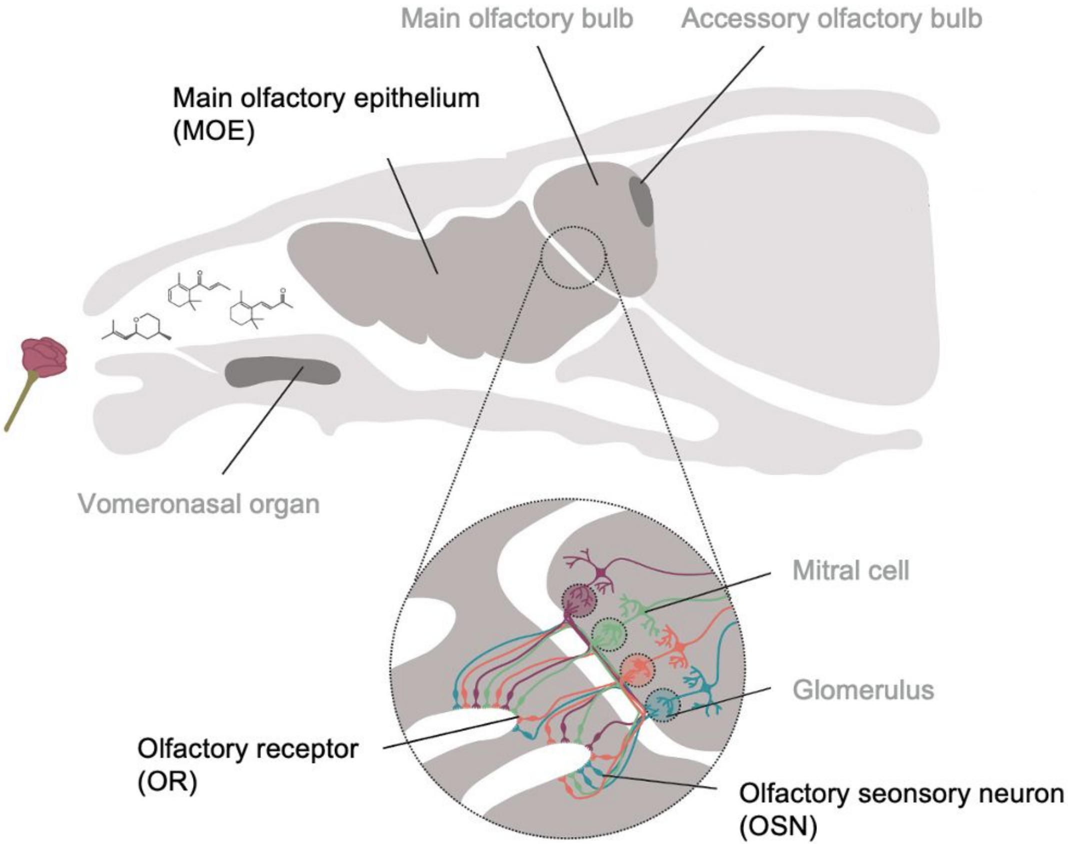
Project goal

Follow the full journey of sequencing data from biological sample to biological insight

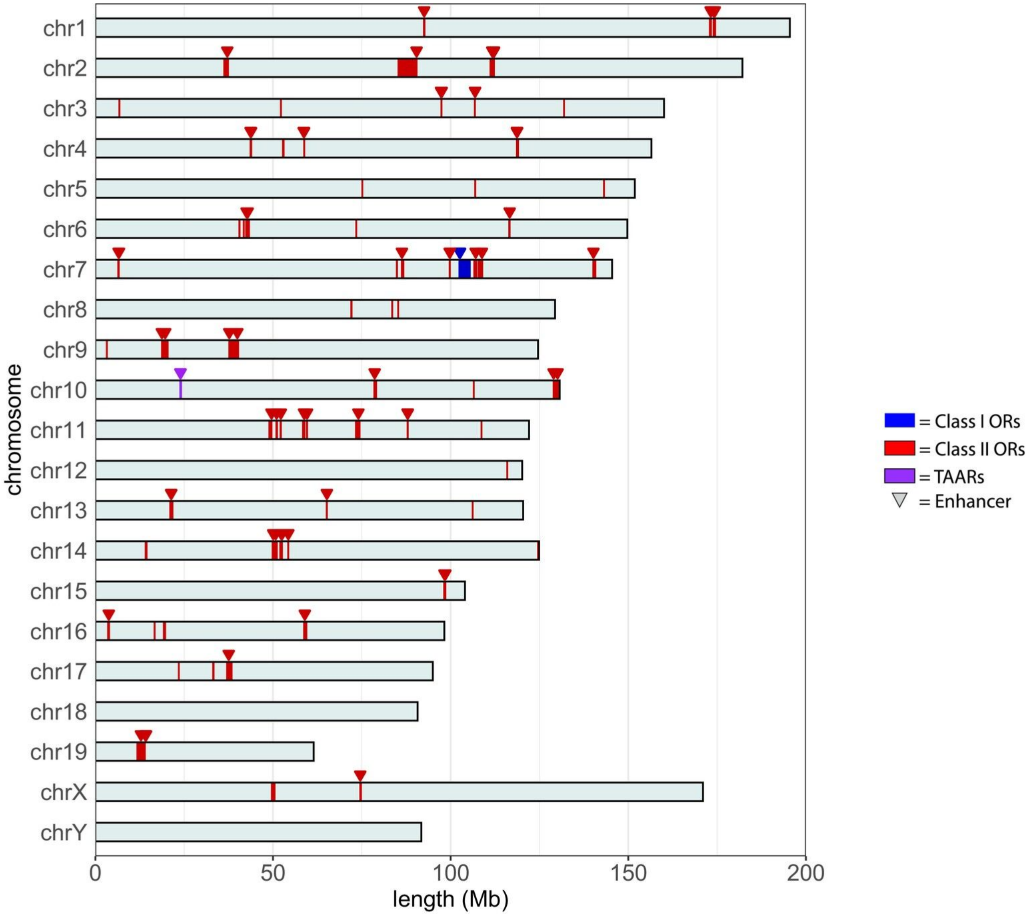
Olfactory system

- Around 1,300 olfactory receptor genes in mice (around 400* in humans)
- Understudied mechanisms of regulation
- My PhD project, so I am biased

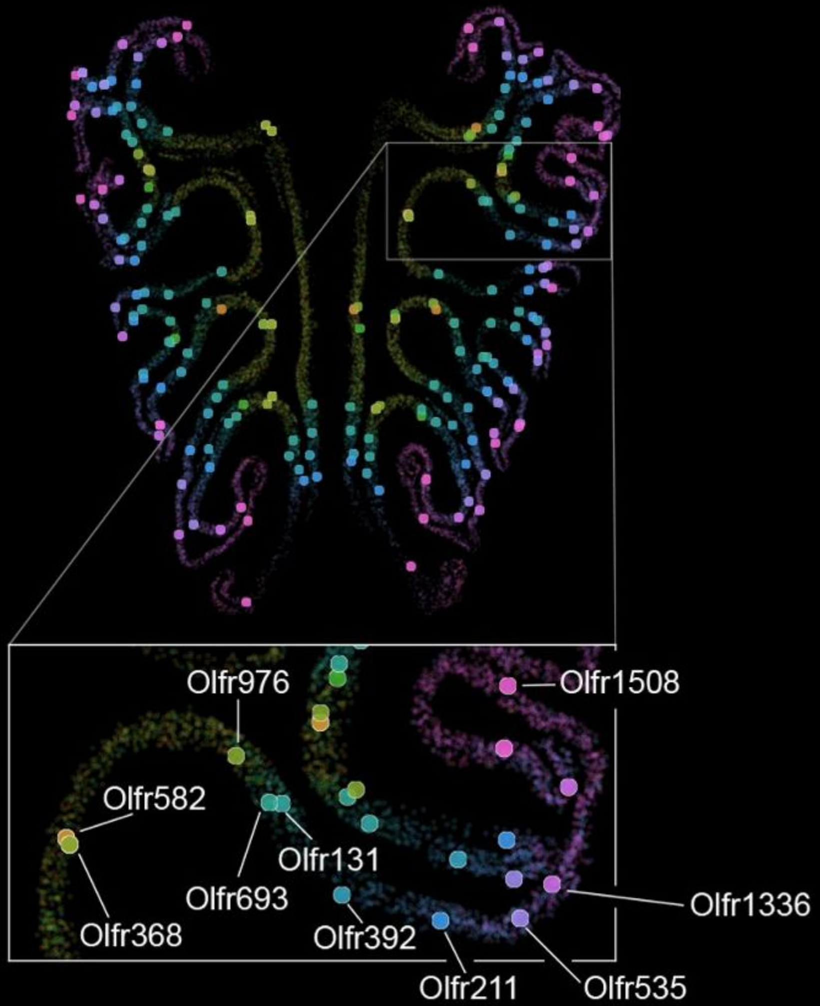


(Adapted from Bear et al., 2016)

Olfactory genes are the largest gene family in mammals



(Yusuf & Monahan, 2024)



https://www.the-scientist.com/a-smell-map-reveals-insights-about-olfactory-organization-74447

Nobel Prize in Physiology / Medicine 2004

Richard Axel and Linda B. Buck

Transcriptomics

The large-scale measurement of **gene expression** — which genes are active and at what level.

Epigenetics

The study of changes in gene activity that are not caused by changes in the DNA sequence itself, but by **chemical modifications of DNA** or **histones**.

How can we approach the system with diverse regulatory mechanisms on the transcriptomic and epigenetic levels?

Next Generation Sequencing (NGS)

A high-throughput technology that “reads” the nucleotide sequence of DNA or RNA fragments. You can profile an entire genome, transcriptome, or epigenome, making it the backbone of modern genomics research.

RNA-seq

A method for measuring gene expression genome-wide by sequencing RNA molecules (normally converted into cDNA) from a cell or tissue.

ChIP-seq

A method for mapping the genomic locations of specific proteins or histone modifications by sequencing DNA that co-precipitates with an antibody target.

Course structure

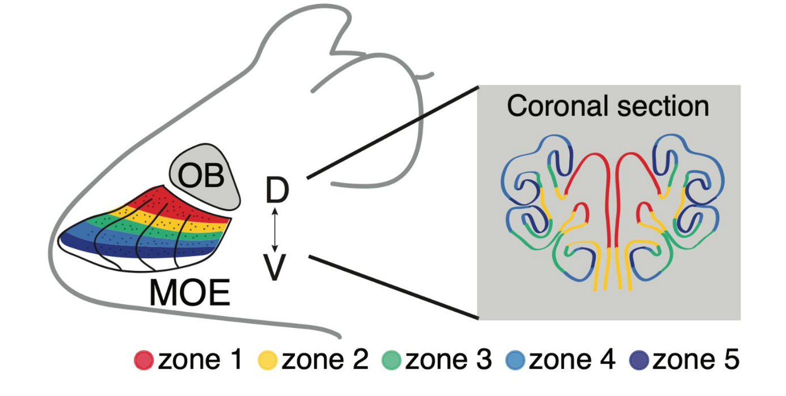
1. Publicly available RNA-seq and ChIP-seq dataset

2. 3 pipelines:
   - Downloading the data
   - RNA-seq
   - ChIP-seq (hopefully)

3. Lecture materials on next-generation sequencing technology and steps that take place before the data analysis

4. Guided tutorials on pipeline execution and analysis roadmap

Publicly available RNA-seq and Chip-seq dataset




eLife

RESEARCH ARTICLE

Opposing, spatially-determined epigenetic forces impose restrictions on stochastic olfactory receptor choice

Elizaveta V Bashkirova[1,2], Nell Klimpert[3], Kevin Monahan[4], Christine E Campbell[5,6], Jason Osinski[5,6], Longzhi Tan[7], Ira Schieren[2], Ariel Pourmorady[1,2], Beka Stecky[2], Gilad Barnea[3], Xiaoliang Sunney Xie[8,9], Ishmail Abdus-Saboor[2], Benjamin M Shykind[10], Bianca J Marlin[2], Richard M Gronostajski[5,6], Alexander Fleischmann[3], Stavros Lomvardas[2,11]*

Pipelines: Snakemake / Nextflow

- **fetchngs** to download the data: https://nf-co.re/fetchngs/1.12.0/
- **3t-seq** for RNA–seq data analysis: https://boulardlab.github.io/3t-seq/
- **nf-core ChIP-seq** for ChIP-seq data analysis: https://nf-co.re/chipseq/2.1.0

By the end of the course you will:

- Understand the main principles behind **NGS technologies** and library preparation
- Be able to set up and run **RNA-seq** and **ChIP-seq** analysis pipelines
- Know how to assess **data quality** and interpret QC reports
- Perform **differential gene expression analysis** and visualise results in **R**
- Understand how chromatin state and gene expression relate to each other
- Have hands-on experience with reproducible computational workflows using **Snakemake** and **Nextflow**

Prerequisites

- Number of people: **ideally up to 5** (we can see what we can do if more people want to join).
- Bash command line: **familiar with** navigating the filesystem, running scripts, and basic syntax from the command line.
- R: **intermediate** - comfortable with data manipulation (**dplyr, tidyverse**), basic plotting (**ggplot2**), and reading/writing files. Experience with **DESeq2** is a plus.

Looking forward to exploring NGS data analysis with you!

:)